

THERMACART · REQUIREMENTS

ThermaCart requirements

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DOCUMENT CONTROL

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Project TRL	TRL 2 Technology concept formulated		
Repository	github.com/BoujeeEnjinia1701/thermacart		
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REVISION HISTORY

Version	Date	Author	Change
0.2	2026-09-25	Amish Chadha	First measurable requirements for TRL 2, with status against the concept estimates
0.1	2026-09-25	Amish Chadha	Initial scaffold

ThermaCart requirements

These are first-pass requirements for the TC-L cartridge and its adapter frame. Targets are proposals for review, awaiting Amish. The status column compares them with the first-order estimates in the design precis (TCT-PRC-001); every estimate will be checked by calculation at TRL 3. Four requirements are **not met** or **at risk** on paper: R3 (C5 grade), R5, R10 and R14.

Table 1. Requirements

ID	Requirement	Target	Verification (TRL 3 or later)	Status at TRL 2
R1	Offer a small set of standard temperature grades	At least three: C5 (melt 4 to 6 °C, cold chain), C25 (24 to 27 °C, comfort), H70 (68 to 72 °C, hot holding)	Material datasheets; later DSC test of each fill batch	Met on paper (commercial PCMs exist for each grade)
R2	Common envelope that fits kitchen and cooler formats	Fits a GN 1/3 slot: at most 325 x 176 mm footprint and 65 mm high, handle included	Model and drawing	Met: 322 x 152 x 61 mm
R3	Enough stored energy per cartridge	60 Wh (216 kJ) or more of usable storage within ± 3 K of the melt point	Calculation per grade	Not met for C5 (about 55 Wh); C25 about 61 Wh (thin margin); H70 about 67 Wh
R4	Carry with one hand	Filled mass 3.0 kg or less	Mass estimate, then weighing	Met: about 2.6 kg
R5	Efficient use of mass	PCM at least 50 % of filled mass	Mass estimate	Not met: about 42 % with the stock 3.18 mm tube
R6	Sealed for life	No leak through 1,000 thermal cycles and a 1 m drop onto concrete in any state	Seal design review at TRL 3; cycle and drop tests belong to TRL 4	Not yet assessed
R7	Long life	1,000 cycles or more with less than 10 % loss of latent capacity	Literature and supplier data at TRL 3	Plausible for paraffins; unassessed for salt hydrates
R8	No pressure hazard	Internal gauge pressure 0.5 bar or less at the highest charge temperature; ullage of at least 10 % of the inner volume at that temperature	Calculation	Plausible (filled hot, estimate); to be calculated
R9	Wrong grade cannot be fitted	Each grade has its own nose key; a frame for one grade rejects the others	Design review of key geometry	Met in concept
R10	Cold-chain hold time	Two C5 cartridges keep a 25 L cooler (heat leak about 0.5 W/K, estimate) at 2 to 8 °C for 8 h or more at 32 °C ambient	Heat balance calculation	At risk: about 8.1 h, no margin
R11	Hot-holding time	One H70 cartridge keeps food in an insulated GN carrier (about 0.25 W/K, estimate) at 63 °C or more for 4 h or more at 25 °C ambient	Heat balance calculation	Met: about 6 h
R12	Recharge with common equipment	Full recharge in 8 h or less: C5 and C25 in a domestic freezer or refrigerator, H70 in an oven at 100 °C or less or on a 60 W heating pad	Calculation (Stefan problem plus fin convection)	At risk for C5 (4 to 8 h, estimate); H70 about 2.5 h
R13	State of charge visible	Charged or melted state readable at a glance without tools	Design review	Met in concept (sight window on a PCM vial)

ID	Requirement	Target	Verification (TRL 3 or later)	Status at TRL 2
R14	Safe to handle when charged	Handle touch temperature below the ISO 13732-1 burn threshold for 10 s contact on every grade	Calculation of handle temperature	Not met for H70 with a bare aluminium handle; insulated grip proposed
R15	Materials compatible with each fill	No corrosion, swelling or permeation: aluminium shell with FKM or NBR seals for paraffins; coated shell or polymer liner for salt hydrates	Compatibility review against supplier data	Met for paraffin grades; open for salt hydrates
R16	Low cost and garage-buildable	\$50 or less in parts per cartridge and \$20 or less per frame; stock sections, drill press and taps only	Priced BOM	Met, thin margin: about \$49 per cartridge, about \$15 per frame (indicative)
R17	Open and documented	Envelope, key geometry, grade table and fill recipe published under CERN-OHL-S-2.0	Repository review	In progress
R18	Recoverable at end of life	Fill can be drained or melted out through the fill port and the shell reused; fill named on the label	Design review	Met in concept

■ Assumptions

- Usable latent storage within the working band: 180 kJ/kg for C5, 200 kJ/kg for C25 and 220 kJ/kg for H70, below the 250, 230 and 260 kJ/kg that Rubitherm quotes for its RT 5 HC, RT 25 HC and RT 70 HC grades, which include sensible heat over a wider band ([Rubitherm](#)).
- Fill of about 1.1 kg: 90 % of a 1.62 L inner volume, filled as liquid at the top of the grade's service temperature, at a liquid density of about 0.77 kg/L (estimate for paraffins).
- Hot-holding threshold of 63 °C follows common food-safety practice (the US FDA Food Code uses 57 °C, 135 °F); the lower value would lengthen the hold time.
- Cooler and carrier heat leaks are estimates for typical polyurethane and EPP boxes and must be replaced with measured values from a partner's real boxes.